

Exp 1 : Transformation between Cartesian, Cylindrical and spherical coordinate system

Aim: To write program for Transformation between Cartesian, Cylindrical and spherical coordinate system using SCILAB.

Software required: SCILAB version 6.0.1

Practical output:

Test case 1:

```
1 clc;
2 x=input('Enter the value of X=');
3 y=input('Enter the value of y=');
4 z=input('Enter the value of z=');
5 r=sqrt(x^2+y^2);
6 pi=atand(y/x);
7 disp('Cartesian to cylindrical coordinate system of p(r,pi,z)=[r pi z]);
8 r=sqrt(x^2+y^2+z^2);
9 theta=acosd(z/r);
10 pi=atand(y/x);
11 disp('Cartesian to spherical coordinate system of s(r,theta,pi)=[r theta pi]);
12
```

Enter the value of X=10

Enter the value of y=15

Enter the value of z=20

"Cartesian to cylindrical coordinate system of p(r,pi,z)="

18.027756 56.309932 20.

"Cartesian to spherical coordinate system of s(r,theta,pi)="

26.925824 42.031114 56.309932

--> |

Test case 2:

```
1 clc;
2 clear;
3 r = input('Enter the value of r = ');
4 phi = input('Enter the value of phi = ');
5 z = input('Enter the value of z = ');
6 x = r * cosd(phi);
7 y = r * sind(phi);
8 disp('Cylindrical to Cartesian Coordinate System of P(x,y,z)=');
9 disp([x y z]);
10 rho = sqrt(r^2 + z^2);
11 theta = acosd(z / rho);
12 disp('Cylindrical to Spherical Coordinate System of S(rho,theta,phi)=');
13 disp([rho theta phi]);
```

Enter the value of r = 10

Enter the value of phi = 15

Enter the value of z = 20

"Cylindrical to Cartesian Coordinate System of P(x,y,z)="

9.6592583 2.5881905 20.

"Cylindrical to Spherical Coordinate System of S(rho,theta,phi)="

22.36068 26.565051 15.

--> |

Test case 3:

```
1 clc;
2 clear;
3 r = input('Enter the value of r = ');
4 phi = input('Enter the value of phi = ');
5 theta = input('Enter the value of theta = ');
6 x = r * sind(theta) * cosd(phi);
7 y = r * sind(theta) * sind(phi);
8 z = r * cosd(theta);
9 disp('Spherical to Cartesian Coordinate System of P(x,y,z)=');
10 disp([x y z]);
11 p = r * sind(theta);
12 z = r * cosd(theta);
13 disp('Spherical to Cylindrical Coordinate System of S(p,phi,z)=');
14 disp([p phi z]);
```

Enter the value of r = 10

Enter the value of phi = 20

Enter the value of theta = 15

"Spherical to Cartesian Coordinate System of P(x,y,z)="

2.4321035 0.8852133 9.6592583

"Spherical to Cylindrical Coordinate System of S(p,phi,z)="

2.5881905 20. 9.6592583

--> |

Theoretical calculations

Test case 1:

i) $x=10$; $y=15$; $z=20$

$$\rho = \sqrt{x^2 + y^2} = \sqrt{10^2 + 15^2} = 18.02$$
$$\phi = \tan^{-1}(y/x) = \tan^{-1}(15/10) = 56.30^\circ$$

$z = 20$

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{10^2 + 15^2 + 20^2} = 26.92$$
$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{20}{26.92}\right) = 42.03^\circ$$
$$\phi = \tan^{-1}(y/x) = \tan^{-1}\left(\frac{15}{10}\right) = 56.30^\circ$$

Test case 2:

ii) $\rho=10$; $\phi=15^\circ$; $z=20$

$$x = \rho \cos \phi = 10 \cos(15^\circ) = 9.65$$
$$y = \rho \sin \phi = 10 \sin(15^\circ) = 2.598$$

$z = z = 20$

$$r = \sqrt{\rho^2 + z^2} = \sqrt{10^2 + 20^2} = 22.36$$
$$\theta = \cos^{-1}\left(\frac{z}{r}\right) = \cos^{-1}\left(\frac{20}{22.36}\right) = 26.565^\circ$$

Test case 3:

iii) $r = 10$; $\theta = 15^\circ$; $\phi = 20$

$$x = r \sin \theta \cos \phi = 10 \cos 20^\circ \sin 15^\circ = 3.303$$
$$y = r \sin \theta \sin \phi = 10 \sin 15^\circ \sin 20^\circ = 0.885$$
$$z = r \cos \theta = 10 \cos 15^\circ = 9.396$$

$$\rho = r \sin \theta = 10 \sin 15^\circ = 3.420$$
$$\phi = \phi = 15^\circ$$
$$z = r \cos \theta = 10 \cos 15^\circ = 9.396$$

0/p2 ✓

Result :

The conversion of coordinates from Cartesian to Cylindrical and Spherical systems and vice versa was successfully performed and verified.